

Element B:**X-RAY Holders****Similar Existing Mechanism #1 < Holders >**

Source: “ Flexi-Holder with Full Motion Tilt, Rotate, and Articulation for CR & DR Plates and Cassettes”
 zzmedical, 2026, [Medical X-ray Holder](#) Accessed 23 Feb. 2026.

Medical X-Ray Holder**Details:**

The Flexi-holder significantly reduces retakes caused by improper cassette positioning. It permits smoother scheduling and superior radiographs saving film, time, and labor for cost effective radiographic procedures. The Flexi-holder features an extra wide base for maximum stability. It allows accurate radiographs to be taken from virtually any angle from floor level to 71”+ (180 cm) high. It rolls into position on large, locking casters and tilts the cassette 180° facing straight up, down, or anywhere in between. The Flexi-holder accept virtually all cassettes and cassettes can be rotated 360 degrees clockwise or counterclockwise. The arm rotates 360 degrees on the vertical column. Great for three-way studies of acute abdomens, myelograms, lateral decubitus fluid level studies of abdomens and chest, operating room radiography, and for fractures where patients are immobilized.

Analysis:

This is the Medical X-ray Holder that the zookeepers showed as an example, however this design had many issues, such as it is very expensive, it is very limited to turning in specific angles, and cannot hang onto a pole for them to conveniently place and take the most efficient X-rays.

ARM Mechanisms

Similar Existing [ARM] Mechanism #1 < TV Mounts >

Source: “Mount-It! TV Mount Full Motion Indoor Wall TV Mount Fits TVs up to 32-in (Hardware Included).” Lowe’s, 2026, [Lowes Mount Link](#) Accessed 2 Mar. 2026.

TV Mount



Details:

A TV pole mount is a lightweight, portable device clamped on to a pole in order to support and angle a TV. The mount relies on friction between the pole surface, in order to support the weight of the TV. TV mounts are common in homes, however the pole mounted versions are more commonly used in commercial settings.

Analysis:

This design would be unsuccessful for our problem because the attachment point was designed to attach to a circular pole with a 1.1 inch to 2.4 inch diameter. The pole we need to attach too is square with 5 inch rounded sides. The Tv mount bracket would also be unsuitable for our X-ray plate as it was designed to screw into the TV, but our X-ray plate has no ability to be screwed into. It needs to be clamped on to the sides as well.

Similar Existing [ARM] Mechanism #2 < Monitor wall mounts >

Source: “MOUNTUP Monitor Wall Mount” MOUNTUP US, MOUNTUP, 2026, <https://a.co/d/0efoYDau>

Accessed 16 Mar. 2026.

MOUNTUP Monitor Wall Mount**Details:**

This heavy-duty monitor wall mount is specifically engineered to support ultrawide and curved screens ranging from 24 to 49 inches. It features a high-performance gas spring system that allows for fluid height adjustments within a 13-inch range and can hold weights between 4.4 and 44 lbs. The mount provides a full range of motion, including $+90^{\circ}/-45^{\circ}$ tilt, $\pm 90^{\circ}$ swivel, and 360° rotation for both landscape and portrait orientations.

**Analysis:**

After thorough searching, we found that this wall monitor arm mount is the best arm to get the X-Ray to move around and be adjusted with. This follows the requirements and suggestions the zookeepers wished for, and fit well with the design we wanted in the end. This monitor arm can move in all 3 dimensions following our criteria and constraints, it can move left and right, tilt the xray plate 360° and move up and down with tension etc.

CLAMP to plate Mechanisms

Similar Existing [CLAMP] Mechanism #1 < Clamps >

Source: "6 Inch Malleable Iron C-Clamp" Tekton, Tekton, 2026, [Screw clamp](#) Accessed 16 Mar. 2026.



Details:

The Tekton 6-inch malleable iron C-clamp (4027) is a heavy-duty, screw-type mechanism designed for high-pressure applications in welding and woodworking. It features a cast iron I-beam frame, Acme-threaded screws, and a swivel pad for uniform pressure and workpiece protection.

Analysis:

This is the clamp design we used as an example/reference so we can make our own clamp to plate mechanism, we needed to get some kind of connection to hold the plate to the rest of the arm monitor/rest of the design so this was an important part to assemble all of our parts in the end.

Other CLAMP to pole Mechanisms

Similar Existing Mechanism #1 < Ratchet straps >

Source: “ Retractable Ratchet Tie-Downs - S-Hook” POWERTYE MFG.com/amazon, amazon, 2026, [Ratchet Straps](#) Accessed 16 Mar. 2026.

Ratchet strap



Details:

Uline Retractable Ratchet Tie-Downs feature push-button, self-retracting straps that meet WSTDA standards for securing loads on light-duty trucks and trailers. Available in various sizes and capacities, these S-hook, vinyl-coated straps include integrated break strength and working load limit information on a sewn-on tag.

Analysis:

We need Ratchet Straps to properly attach the X-ray holder device to the 5” by 5” pole as the zookeepers wished for. This is essential for keeping our device on the pole for a long time and keep it everything stable/together. The zookeepers can move the X-ray up and down along the pole to get their desired pictures the way they want allowing more variety with the pictures.

Latch Mechanisms

Similar Existing Mechanism #1 < Latches >

Source: “Matte Black Chest Door Latch ” homedepot.com, home depot, 2026, [Door Latches](#) Accessed 16 Mar. 2026.

Door Latches



Details:

The Everbilt (2-Pack) 2-3/4 in. x 1-1/2 in. Matte Black Chest Door Latch is a surface-mounted steel hardware solution suited for light-duty decorative chests, storage boxes, and DIY projects. While users praise the product's easy installation and aesthetic appeal, reviewers note the metal is thin and the latch relies on light friction rather than a heavy-duty mechanism.

Analysis:

We need latches to get our X-ray plate in its case to stay in a second casing we designed to attach to the monitor arm and the rest of our device. This should prevent the X-ray from falling out of the device/boxing when the zookeepers want to tilt the X-ray holder to get specific shots/X-rays of the giraffes or other animals. Tiling the X-Ray is essential for good quality scans and helps with getting data to keep animals healthy and up to date on their health/etc. This could prevent risks and detect issues later in the futur for the animal’s safety, therefore designing a X-Ray holder with good protection/locks can help keep these X-Rays intact and prevent damage from falling on the ground.

References

1. “Mount-It! TV Mount Full Motion Indoor Wall TV Mount Fits TVs up to 32-in (Hardware Included).” Lowe’s, 2026, [Lowes Mount Link](#) Accessed 2 Mar. 2026.
2. “Single Monitor Arm Wall Mount | LX pro | Ergotron.” Ergotron.com, Ergotron, 2026, www.ergotron.com/en-us/products/product-details/45-683#?color=dark%20grey&buynow=0. Accessed 5 Mar. 2026.
3. “MOUNTUP Monitor Wall Mount for 49 45 43 40 35 32 27 24 Inch Screens, Heavy Duty Wall Monitor Mount, Holds 4.4 to 44 lbs, Ultrawide Monitors Arm Wall Mount, VESA Mount 75x75/100x100mm ” MOUNTUP US, MOUNTUP, 2026, <https://a.co/d/0efoYDau> Accessed 16 Mar. 2026.
4. “6 Inch Malleable Iron C-Clamp” Tekton, Tekton, 2026, [Screw clamp](#) Accessed 16 Mar. 2026.
5. “Uline Retractable Ratchet Tie-Downs - S-Hook” ULINE.com, ULINE, 2026, [Ratchet Straps](#) Accessed 16 Mar. 2026.
6. “Matte Black Chest Door Latch ” homedepot.com, home depot, 2026, [Door Latches](#) Accessed 16 Mar. 2026.